

PROJECT REPORT

HR Employee Attrition Analysis

Identifying Key Drivers of Employee Turnover to Reduce Hiring Costs

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Tools: MySQL • Power BI • DAX

Dataset: IBM HR Analytics — Kaggle | 1,470 Records | 35 Features

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1. INTRODUCTION

1.1 Background

Employee attrition — the gradual loss of employees over time — is one of the most costly and disruptive challenges organizations face. When an employee leaves, a company does not simply lose a salary; it incurs a chain of hidden costs including recruitment fees, lost productivity during vacancy, onboarding time, and the gradual erosion of institutional knowledge.

Studies by SHRM and Gallup estimate that replacing a single employee costs between 50% to 200% of their annual salary, depending on seniority. For organizations with high attrition rates, this translates into millions of dollars in annual replacement costs — most of which go untracked.

1.2 Objective

This project analyzes IBM's HR dataset to answer three core business questions:

- Why are employees leaving? — Identifying key drivers such as compensation, overtime, and job satisfaction.
- Which departments and roles are at highest risk? — Pinpointing where attrition is most concentrated.
- What is the financial impact? — Quantifying the estimated cost of attrition per department.

The goal is to provide HR analysts with a self-service Power BI dashboard that surfaces these insights interactively — enabling data-driven retention decisions before talent walks out the door.

1.3 Dataset Overview

- Source: IBM HR Analytics Employee Attrition dataset — available on Kaggle
- Records: 1,470 employees
- Features: 35 columns covering demographics, compensation, satisfaction ratings, and work history
- Target Variable: Attrition (Yes = Left, No = Stayed)
- Missing Values: None — clean dataset ready for analysis

2. TECHNOLOGY STACK

This project follows a two-stage pipeline: data storage and transformation in SQL, and interactive visualization in Power BI.

Tool	Version / Type	Purpose
MySQL 8.0	Relational Database	Data storage, cleaning, transformation, and analytical queries
MySQL Workbench	GUI Client	Schema design, CSV import, and SQL query execution
Power BI Desktop	BI Tool	Dashboard design, DAX measures, drill-through, and RLS
DAX	Query Language	Custom measures including Attrition Rate, Cost, and High Risk Count
GitHub	Version Control	Project documentation, SQL scripts, and notebook hosting

3. PROJECT PROCESS

Step 1 — Data Acquisition

Downloaded the IBM HR Analytics Employee Attrition dataset from Kaggle. The dataset contains 1,470 employee records with 35 features covering age, department, job role, monthly income, overtime status, job satisfaction, years at company, and attrition status.

Step 2 — Database Setup (MySQL)

Created a dedicated database and table in MySQL to store the dataset:

```
CREATE DATABASE hr_attrition;
USE hr_attrition;

CREATE TABLE employee (
```

```

    EmployeeNumber INT PRIMARY KEY,
    Age INT, Attrition VARCHAR(5),
    Department VARCHAR(50), JobRole VARCHAR(50),
    MonthlyIncome INT, OverTime VARCHAR(5),
    JobSatisfaction INT, YearsAtCompany INT,
    TotalWorkingYears INT, ... -- 35 columns total
);

LOAD DATA INFILE 'path/to/file.csv'
INTO TABLE employee
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

```

Three redundant columns — EmployeeCount, Over18, and StandardHours — were retained in the schema but excluded from all analysis as every record had identical values (1, Y, and 80 respectively).

Step 3 — SQL Analysis & Transformation

Seven analytical queries were written to explore attrition patterns before building the dashboard:

Query 1 — Overall Attrition Rate

```

SELECT
    COUNT(*) AS TotalEmployees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS TotalAttrition,
    ROUND(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END)
          * 100.0 / COUNT(*), 2) AS AttritionRate
FROM employee;
-- Result: 1470 employees, 237 left, 16.12% attrition rate

```

Query 2 — Attrition by Department

```

SELECT Department,
    COUNT(*) AS TotalEmployees,
    SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS Attritions,
    ROUND(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END)
          * 100.0 / COUNT(*), 2) AS AttritionRate
FROM employee
GROUP BY Department
ORDER BY AttritionRate DESC;
-- Result: Sales 20.63%, HR 19.05%, R&D 13.84%

```

Query 3 — Attrition by Age Band

```

SELECT
    CASE
        WHEN Age < 25 THEN 'Under 25'
        WHEN Age BETWEEN 25 AND 34 THEN '25-34'
        WHEN Age BETWEEN 35 AND 44 THEN '35-44'
        WHEN Age BETWEEN 45 AND 54 THEN '45-54'
        ELSE '55+'
    END AS AgeBand,
    ROUND(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END)

```

```

        * 100.0 / COUNT(*), 2) AS AttritionRate
FROM employee
GROUP BY AgeBand
ORDER BY AttritionRate DESC;
-- Result: Under 25 = 39.18% (highest risk)

```

Query 4 — Salary: Leavers vs Stayers

```

SELECT Attrition,
       ROUND(AVG(MonthlyIncome), 2) AS AvgMonthlyIncome
FROM employee
GROUP BY Attrition;
-- Result: Leavers avg $4,787 vs Stayers avg $6,833

```

Query 5 — Overtime Impact

```

SELECT OverTime,
       ROUND(SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END)
             * 100.0 / COUNT(*), 2) AS AttritionRate
FROM employee
GROUP BY OverTime;
-- Result: No OT = 10.44%, Yes OT = 30.53%

```

Query 6 — High Risk Profile

```

SELECT JobRole, COUNT(*) AS HighRiskCount
FROM employee
WHERE Attrition = 'Yes'
      AND OverTime = 'Yes'
      AND JobSatisfaction <= 2
      AND MonthlyIncome < (SELECT AVG(MonthlyIncome) FROM employee)
GROUP BY JobRole
ORDER BY HighRiskCount DESC;
-- Result: Research Scientist (15), Lab Technician (12), Sales Rep (7)

```

Query 7 — Estimated Attrition Cost by Department

```

SELECT Department,
       SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS EmployeesLost,
       ROUND(SUM(CASE WHEN Attrition = 'Yes'
                 THEN MonthlyIncome END) * 12 * 0.5, 2) AS EstimatedAttritionCost
FROM employee
GROUP BY Department
ORDER BY EstimatedAttritionCost DESC;
-- Using 50% of annual salary as replacement cost benchmark
-- R&D: $3.28M, Sales: $3.26M, HR: $0.27M

```

Step 5 — Power BI Dashboard

Connected Power BI Desktop directly to MySQL (localhost / hr_attrition). Removed redundant columns in Power Query. Created 5 DAX measures in a dedicated _Measures table before building any visuals.

4. KEY DAX MEASURES

All measures were created in a dedicated `_Measures` table to keep the data model clean and organized.

Measure	DAX Formula	Purpose
Total Employees	<code>COUNTROWS (employee)</code>	Headcount for any filter context
Total Attrition	<code>CALCULATE (COUNTROWS (employee), employee[Attrition] = "Yes")</code>	Count of employees who left
Attrition Rate	<code>DIVIDE([Total Attrition], [Total Employees], 0)</code>	Percentage formatted as %
Avg Monthly Income	<code>AVERAGE (employee[MonthlyIncome])</code>	Average salary in current context
Attrition Cost	<code>CALCULATE (SUMX (employee, employee[MonthlyIncome] * 12 * 0.5), employee[Attrition] = "Yes")</code>	Estimated replacement cost at 50% of annual salary
Avg Tenure Leavers	<code>CALCULATE (AVERAGE (employee[YearsAtCompany]), employee[Attrition] = "Yes")</code>	Avg years before resignation
High Risk Count	<code>VAR AvgInc = CALCULATE (AVERAGE (employee[MonthlyIncome]), ALL (employee)) RETURN CALCULATE (COUNTROWS (employee), Attrition="Yes", OverTime="Yes", JobSatisfaction<=2, MonthlyIncome<AvgInc)</code>	Employees meeting all 4 high-risk criteria
Attrition Rate No OT	<code>CALCULATE ([Attrition Rate], employee[OverTime] = "No")</code>	Rate for non-overtime employees
Attrition Rate Yes OT	<code>CALCULATE ([Attrition Rate], employee[OverTime] = "Yes")</code>	Rate for overtime employees

5. DASHBOARD STRUCTURE

The dashboard is built across three pages, each serving a distinct analytical purpose:

Page 1 — Executive Summary

Designed for HR managers who need a quick overview of the attrition situation across the organization.

- 5 KPI Cards: Attrition Cost (\$6.81M), Attrition Rate (16.12%), Avg Monthly Income (\$6.50K), Total Attrition (237), Total Employees (1,470)
- Attrition Rate by Department (Horizontal Bar) — Sales 20.63%, HR 19.05%, R&D 13.84%
- Attrition by Overtime — Donut chart filtered to leavers only. 53.59% of leavers were doing overtime
- Attrition Rate by Age Group (Column Chart) — Under 25 at 39.18% is the highest risk group
- High Risk Profiles (Bar Chart) — Research Scientists and Lab Technicians identified as highest risk roles
- Slicers: Gender, Marital Status, Education Field — all visuals respond dynamically to slicer selections

Page 2 — Deep Dive Analysis

Designed for data analysts and HR business partners who want to explore multi-variable relationships.

- Scatter Plot — Income vs Tenure coloured by Attrition: Shows leavers (red) clustered in the low income, low tenure zone
- Clustered Bar — Headcount vs Attrition by Job Role: Sales Representatives have the highest leaver-to-headcount ratio
- Matrix Table — Attrition Rate by Department × Overtime: Overtime employees leave at 2–3x the rate across all departments
- Attrition Cost by Department (Bar Chart): R&D (\$3.28M) and Sales (\$3.26M) account for ~96% of total replacement cost
- Supporting Cards: Avg Tenure (5.13 Years), Avg Job Satisfaction (2.73/4), Avg Work Life Balance (2.76/4)

Page 3 — Department Drill Through

Accessed by right-clicking any department on Page 1 and selecting Drill Through. The entire page auto-filters to the selected department.

- 4 KPI Cards: Department-specific Total Employees, Total Attrition, Attrition Rate, Attrition Cost
- Attrition Rate by Job Role — shows which roles within the department are highest risk
- Attrition Rate by Age Group — department-level age band breakdown
- Leavers by Job Satisfaction Level — shows whether leavers were satisfied or dissatisfied
- Employee Leaver Details Table — individual records of each employee who left, showing salary, tenure, overtime, satisfaction, and job level

Advanced Features Implemented

- **Drill Through:** Right-click any department bar → drill to department-specific detail page with auto back navigation
- **Custom Tooltips:** Hover over any chart element to see enriched context — Total Employees, Attrition Rate, Avg Income, and Attrition Cost
- **Row Level Security (RLS):** Roles defined for Department managers so each sees only their own department's data when the report is published
- **Dynamic Slicers:** Gender, Marital Status, and Education Field slicers filter all Page 1 visuals simultaneously
- **AgeBand Sort Column:** Custom sort column (AgeBandSort) created to display age bands in logical order rather than alphabetical

6. KEY FINDINGS & ANALYSIS

16.12% Attrition Rate Above industry avg of 10–12%	237 Total Leavers Out of 1,470 employees	\$6.81M Est. Annual Cost 50% of annual salary benchmark
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Sales Highest Risk Dept 20.63% attrition rate	Sales Rep Highest Risk Role 39.76% attrition rate	5.13 yrs Avg Tenure (Leavers) Before resignation
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6.1 Detailed Findings

Finding	Data	Business Implication
Overall attrition is above benchmark	16.12% vs 10–12% industry average	Every 6th employee is leaving — significant drain on recruitment budget
Sales is the highest risk department	20.63% attrition — nearly 1 in 5 Sales employees left	High client-facing turnover disrupts relationships and pipeline
Young employees leave the most	Under 25: 39.18% — over 1 in 3 leave	Early career hires need structured growth paths and competitive pay
Compensation drives attrition	Leavers earn avg \$4,787/mo vs \$6,833/mo for stayers	A \$2,000/month income gap is significant enough to drive job-switching
Overtime doubles attrition risk	10.44% without OT vs 30.53% with OT	Burnout is consistent across all departments — workload policy needs review
Research Scientists & Lab Techs are highest risk profiles	15 and 12 employees match all 4 high-risk criteria	Skilled technical roles being lost to better-paying competitors
R&D and Sales bear most replacement cost	\$3.28M and \$3.26M respectively	Despite different attrition rates, high headcount in R&D equals high total cost
Avg tenure of leavers is only 5.13 years	Employees leave before reaching seniority	Loss of institutional knowledge and training investment

7. BUSINESS RECOMMENDATIONS

Based on the analysis, the following actions are recommended to reduce attrition and cut hiring costs:

1. Review Compensation for High-Risk Roles

The \$2,000/month income gap between leavers and stayers is the single clearest driver in this dataset. Research Scientists and Laboratory Technicians — who have high skill requirements but relatively low salaries (\$2.8K–\$3.3K) — are leaving at disproportionate rates. A targeted salary review for these roles could prevent the most expensive attrition.

2. Address Overtime Policies

Employees working overtime leave at 30.53% vs 10.44% for those who don't — nearly triple the rate. This pattern is consistent across all three departments, making burnout a systemic issue rather than a department-specific one. Reviewing

workload distribution, hiring to reduce overtime dependency, or compensating overtime more competitively would directly reduce attrition.

3. Focus Retention on Under-25 Employees

At 39.18%, the under-25 age group has the highest attrition rate. This cohort is most likely to job-hop for better salaries or growth opportunities in the first few years. Structured mentorship programs, clear career paths, and competitive entry-level pay would improve early-career retention.

4. Prioritize Sales Department Retention

Sales has the highest attrition rate (20.63%) and the second highest replacement cost (\$3.26M). High turnover in a client-facing department also carries indirect costs — lost client relationships, interrupted pipelines, and reputational risk. Incentive restructuring and workload management in Sales should be a near-term priority.

8. CONCLUSION

This project demonstrates a complete end-to-end BI workflow — from raw CSV data in MySQL to an interactive multi-page Power BI dashboard with advanced features including drill-through, custom tooltips, dynamic slicers, and row-level security.

The analysis reveals that IBM's HR attrition is primarily driven by three compounding factors: below-market compensation, excessive overtime, and insufficient growth opportunities for young professionals. Together, these factors are estimated to cost the organization \$6.81M annually in replacement costs — a number that could be significantly reduced through targeted, data-driven HR interventions.

The dashboard is designed to be self-service — HR managers can filter by gender, marital status, or education field, drill into any department for granular detail, and identify individual high-risk employees before they resign.

9. PROJECT FILES & LINKS

- **GitHub Repository:** github.com/RohanParkar — HR-Employee-Attrition-Analysis
- **Power BI File:** `HR_Employee_Attrition_Analysis.pbix`
- **SQL Scripts:** `hr_attrition_schema.sql`, `hr_attrition_queries.sql`
- **Dataset:** IBM HR Analytics — Kaggle (`WA_Fn-UseC_-HR-Employee-Attrition.csv`)